

HEALTHCARE AI

MedBridge

AI Prior Authorization Agent

Prior auth handled in minutes. Not days.

\$19.7B

Annual admin burden

6 tools

End-to-end workflow

Seconds

vs. 3-5 days manual

THE PROBLEM

Healthcare runs on fax machines

Every MRI, surgery, or referral requires prior authorization. Today, that process is deeply broken.

- ! Staff manually fills 10+ page forms and faxes them to insurers — yes, still fax
- ! Payers take 3-5 business days to respond, leaving patients untreated
- ! 1 in 5 requests is denied, triggering hours of additional manual appeal work
- ! \$19.7B lost annually — not because the process is hard, but because humans are doing AI's job

THE SOLUTION

One form. The agent handles everything else.

MedBridge automates the entire prior authorization workflow — from policy scanning to appeal drafting.

1

Scans payer policies and coverage criteria automatically

2

Analyzes patient medical records for medical necessity

3

Auto-fills authorization forms with 96% completeness

4

Submits via EDI 278 to payer portal, monitors in real time

5

If denied — auto-drafts a CMS-compliant clinical appeal

6

HIPAA-aware. Supports Aetna, BCBS, UHC, Cigna, Humana

HOW IT WORKS

Real agentic tool-use loop

The LLM decides which tool to call next, observes the result, and continues — no hardcoded sequences.

1 **scan_insurance_policy** Pulls coverage criteria and clinical requirements from payer database

2 **analyze_medical_records** Evaluates patient history and establishes medical necessity

3 **fill_authorization_form** Auto-populates all required fields with 96% completeness

4 **submit_authorization** Transmits via EDI 278 to the payer portal

5 **track_authorization_status** Checks initial decision, sets up real-time monitoring

6 **draft_appeal** If denied: reads denial code, generates clinical appeal with evidence

KEY DIFFERENTIATOR

Denial auto-appeal — MedBridge doesn't stop at "no"

When payers deny a request, most systems stop. MedBridge immediately fires the appeal workflow.

What happens on denial

- Reads the denial reason code (e.g. CO-50 — medical necessity not established)
- Pulls supporting clinical evidence from the patient record
- Generates a CMS-compliant appeal letter with office notes, VAS scores, failed treatments
- Queues for physician co-signature and auto-submission

Staff time for manual appeal:

2-3 days

-->

seconds

LIVE DEMO

See it in action

Patient

Margaret R. Thompson

Treatment

Lumbar Spine MRI

Insurer

Aetna

Diagnosis

ICD-10 M72.5

Scenario 1 — Standard Approval

Click "Try Demo" to auto-fill, select Standard Approval, watch the agent work all 6 steps.

Authorization confirmed

Scenario 2 — Denial + Auto-Appeal

Agent hits denial CO-50, then immediately drafts a clinical appeal with evidence.

Appeal generated automatically

github.com/kneha07/MedBridge-AI-Prior-Authorization-Agent

ARCHITECTURE

Built on a fully generic agentic platform

React + Vite Frontend

Real-time step polling every 1.5s · Streaming step updates

V

FastAPI Backend · Python 3.11

POST /api/workflow/submit · GET /api/workflow/{id}/status

V

Groq · LLaMA 3.3-70B · Gemini

LLM -> tool_use -> observe -> repeat — real tool loop, not chatbot

V

6 HIPAA-Aware Tools · EDI 278 · CMS 2026

Aetna · BCBS · UHC · Cigna · Humana

MARKET OPPORTUNITY

Why this, why now

\$19.7B

Annual admin burden for prior auth (AMA, 2023)

Jan 2026

CMS mandated prior auth API standards — regulatory tailwind

40%

Physicians report auth delays cause serious patient harm

Zero

Fully automated end-to-end solutions exist today

MedBridge is the agent layer on top of existing payer portals — no replacement of infrastructure required.

ROADMAP

Vertical platform — one Python module = one new work

The agent loop, frontend, and polling are fully generic. Healthcare is just the first vertical.

V	Prior authorization — live today Full end-to-end with denial auto-appeal	LIVE
+	Step therapy exceptions Override fail-first drug requirements with clinical evidence	NEXT
+	Specialist referrals Automate HMO referral submissions and approvals	NEXT
+	Medication appeals Dispute formulary denials for non-covered drugs	NEXT

MEDBRIDGE

Prior auth in minutes. Not days.

Real agentic tool-use · EDI 278 · HIPAA-aware · CMS 2026 compliant

github.com/kneha07/MedBridge-AI-Prior-Authorization-Agent

\$19.7B

problem size

6 tools

agentic workflow

seconds

not 3-5 days